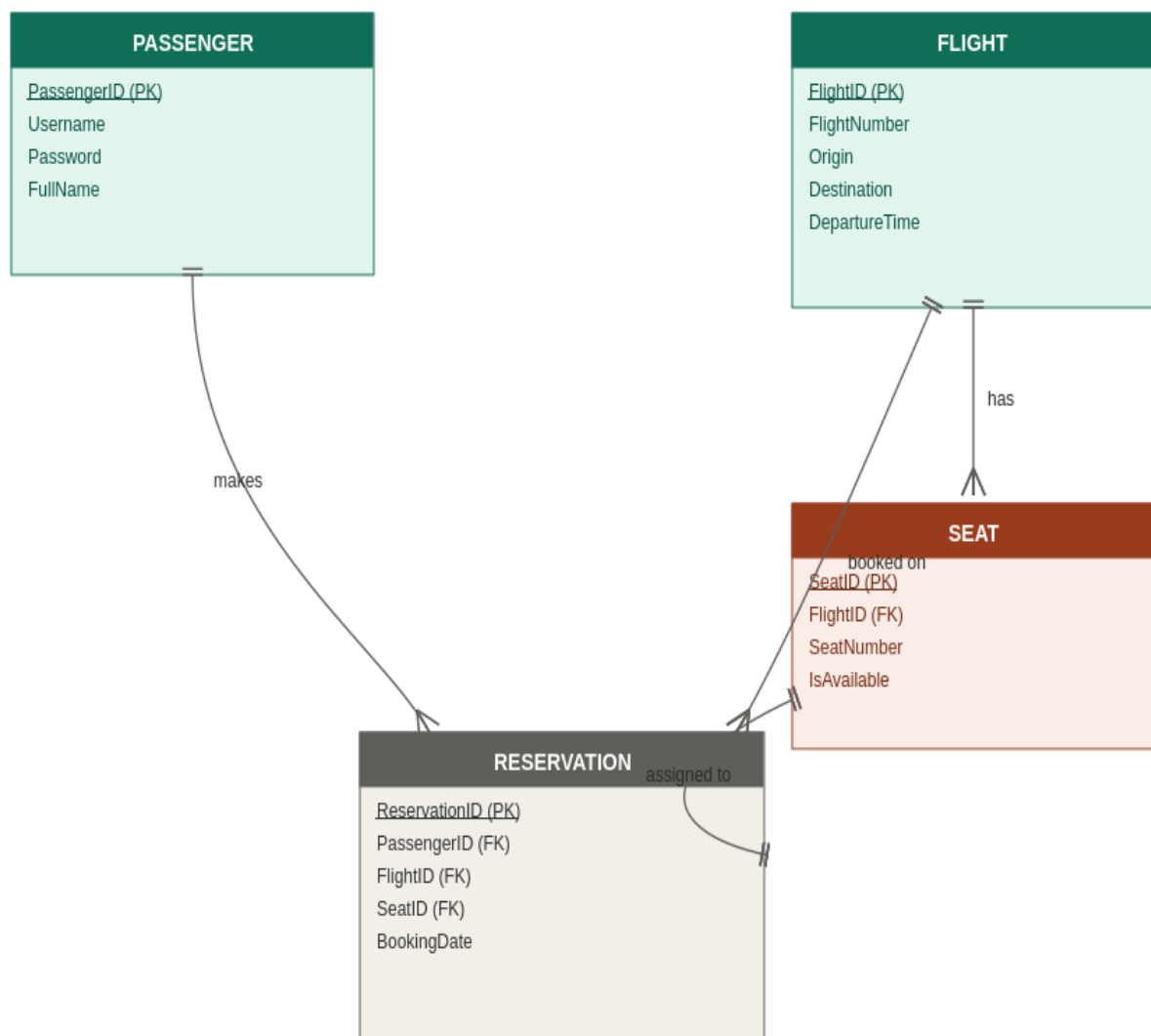


Entity relationship diagram

The ERD models the underlying data for the system: passengers, flights, seats, and reservations. A reservation is the associative entity that ties a passenger, a flight, and a specific seat together, which is what a database schema for this system would need even though the current code keeps this data in memory rather than in tables.



Entities and key attributes

Entity	Key attributes
PASSENGER	PassengerID (PK), Username, Password, FullName
FLIGHT	FlightID (PK), FlightNumber, Origin, Destination, DepartureTime
SEAT	SeatID (PK), FlightID (FK), SeatNumber, IsAvailable
RESERVATION	ReservationID (PK), PassengerID (FK), FlightID (FK), SeatID (FK), BookingDate

Relationships

Relationship	Cardinality	Meaning
PASSENGER makes RESERVATION	1 to many	A passenger can make many reservations; each reservation belongs to exactly one passenger.
FLIGHT has SEAT	1 to many	A flight has many seats; each seat belongs to exactly one flight.
FLIGHT booked on RESERVATION	1 to many	A flight can appear on many reservations; each reservation is for exactly one flight.
SEAT assigned to RESERVATION	1 to 1	A seat is assigned to at most one active reservation, and each reservation holds exactly one seat.

Notes

- PassengerID, FlightID, and SeatID in RESERVATION are foreign keys referencing the primary keys of PASSENGER, FLIGHT, and SEAT respectively.
- SEAT.FlightID is also a foreign key, since every seat belongs to exactly one flight.
- IsAvailable on SEAT can be derived from whether an active RESERVATION currently references that seat, so some schema designs drop this column in favor of checking RESERVATION directly.